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DISPLAY DEVICE IN WHICH PAD ELECTRODES OF A CIRCUIT BOARD AND PAD ELECTRODES OF A DISPLAY PANEL ARE ALIGNED AND A METHOD FOR FABRICATING THE SAME

Abstract

A display device includes: a display panel; and a circuit board connected to the display panel, wherein the circuit board includes a plurality of sub-bonding portions that are separated from each by a plurality of slits, and a length of each of the plurality of sub-bonding portions is set based on an elongation rate of the circuit board.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 U.S.C. §119 to Korean Patent Application No. 10-2024-0027487, filed on Feb. 26, 2024 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] Embodiments of the present invention relates to a display device, and more particularly, to a display device in which pad electrodes of a circuit board and pad electrodes of a display panel are aligned, and a method for fabricating the same.

DISCUSSION OF THE RELATED ART

[0003] Recently, flat panel display devices such as organic light emitting diode display devices have been widely used in various electronic devices such as televisions, laptop computers, and smart phones.

[0004] In many cases, a flat display device generally includes a display panel including a substrate made of glass, and a flexible circuit board may be connected to the display panel.

SUMMARY

[0005] According to an embodiment of the present invention, a display device includes: a display panel; and a circuit board connected to the display panel, wherein the circuit board includes a plurality of sub-bonding portions that are separated from each by a plurality of slits, and a length of each of the plurality of sub-bonding portions is set based on an elongation rate of the circuit board.

[0006] In an embodiment of the present invention, the length of the sub-bonding portion is calculated by a following equation: $C=B/A$, where A is an elongation rate of the sub-bonding portion per unit length, B is a maximum elongation amount of the sub-bonding portion, and C is the length of the sub-bonding portion.

[0007] In an embodiment of the present invention, B is set within 50% of a pitch of an outermost pad electrode of a plurality of pad electrodes that are disposed on the sub-bonding portion.

[0008] In an embodiment of the present invention, the circuit board includes: an insulating layer; a pad electrode disposed on the insulating layer; a base layer disposed on the pad electrode; an adhesive layer disposed on the base layer; and a cover layer disposed on the adhesive layer.

[0009] In an embodiment of the present invention, the base layer includes the plurality of slits.

[0010] In an embodiment of the present invention, the base layer includes the plurality of slits in a bonding area of the circuit board.

[0011] In an embodiment of the present invention, the plurality of slits are arranged in the base layer along a first direction, each of the plurality of slits extends in a second direction, and the first direction and the second direction cross each other.

[0012] In an embodiment of the present invention, the base layer includes a polyimide.

[0013] In an embodiment of the present invention, the insulating layer includes a same material as the base layer.

[0014] In an embodiment of the present invention, the display device further includes an anisotropic conductive film disposed between the display panel and the circuit board.

[0015] In an embodiment of the present invention, the insulating layer is disposed in a non-bonding area of the circuit board.

[0016] In an embodiment of the present invention, the pad electrode is disposed in a non-bonding area and a bonding area of the circuit board.

[0017] In an embodiment of the present invention, the base layer is disposed in a non-bonding area and a bonding area of the circuit board.

[0018] In an embodiment of the present invention, the adhesive layer is disposed in a non-bonding area and a bonding area of the circuit board.

[0019] In an embodiment of the present invention, the cover layer is disposed in a non-bonding area and a bonding area of the circuit board.

[0020] In an embodiment of the present invention, the pad electrode of the circuit board is connected to a pad electrode of the display panel.

[0021] In an embodiment of the present invention, the pad electrode of the circuit board is connected to the pad electrode of the display panel in a bonding area of the circuit board.

[0022] According to an embodiment of the present invention, a method for fabricating a display device includes: disposing an anisotropic conductive film on a substrate of a display panel; disposing a circuit board on the anisotropic conductive film, wherein the circuit board includes a plurality of sub-bonding portions that are separated from each other by a plurality of slits; disposing a heat conduction sheet on the circuit board; disposing a heating/pressurizing header on the heat conduction sheet; and connecting a pad electrode of the circuit board and a pad electrode that is disposed on the substrate to each other by pressurizing and heating the heat conduction sheet with the heating/pressurizing header, wherein a length of each of the plurality of sub-bonding portions is set based on an elongation rate of the circuit board.

[0023] In an embodiment of the present invention, the length of the sub-bonding portion is calculated by a following equation: $C=B/A$, where A is an elongation rate of the sub-bonding portion per unit length, B is a maximum elongation amount of the sub-bonding portion, and C is the length of the sub-bonding portion.

[0024] In an embodiment of the present invention, B is set within 50% of a pitch of an outermost pad electrode of a plurality of pad electrodes that are disposed on the sub-bonding portion, wherein the plurality of pad electrodes includes the pad electrode of the circuit board.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The above and other aspects and features of the present invention will become more apparent by describing in detail embodiments thereof with reference to the attached drawings, in which:

[0026] FIG. 1 is a perspective view of a display device according to an embodiment of the present invention;

[0027] FIG. 2 is an exploded perspective view of a circuit board according to an embodiment of the present invention;

[0028] FIG. 3 is a cross-sectional view taken along line X1-X1' of FIG. 2;

[0029] FIG. 4 is a cross-sectional view taken along line X2-X2' of FIG. 2;

[0030] FIG. 5 is a cross-sectional view taken along line X3-X3' of FIG. 2; and

[0031] FIGS. 6, 7 and 8 are cross-sectional views for describing processes of a method for fabricating the display device according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0032] Features of the present invention and methods to achieve them will become apparent from the descriptions of embodiments hereinbelow with reference to the accompanying drawings.

However, the present invention is not limited to embodiments disclosed herein but may be implemented in various different ways.

[0033] As used herein, a phrase “an element A on an element B” refers to that the element A may be disposed directly on the element B and/or the element A may be disposed indirectly on the element B with another element C interposed therebetween. Like reference numerals may denote like elements throughout the specification and drawings. The figures, dimensions, ratios, angles, numbers of elements given in the drawings are merely illustrative and are not limiting.

[0034] It will be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. For instance, a first element discussed

below could be termed a second element without departing from the spirit and scope of the present invention. Similarly, the second element could also be termed the first element.

[0035] Features of each of various embodiments of the present invention may be partially or entirely combined with each other and may technically variously interwork with each other, and respective embodiments may be implemented independently of each other or may be implemented together in association with each other.

[0036] Hereinafter, embodiments of the present invention will be described with reference to the accompanying drawings.

[0037] FIG. 1 is a perspective view of a display device according to an embodiment of the present invention.

[0038] Referring to FIG. 1, a display device **10** may be applied to portable electronic devices such as mobile phones, smartphones, tablet personal computers (PCs), mobile communication terminals, electronic notebooks, electronic books, portable multimedia players (PMPs), navigation devices, and ultra mobile PCs (UMPCs). As an example, the display device **10** may be applied as a display unit of televisions, laptop computers, monitors, billboards, or the Internet of Things (IOTs). As another example, the display device **10** may be applied to wearable devices such as smart watches, watch phones, glasses-type displays, and head mounted displays (HMDs).

[0039] The display device **10** may have a shape similar to a rectangular shape in a plan view. For example, the display device **10** may have a shape similar to a rectangular shape, in plan view, having long sides in a first direction DR1 and short sides in a second direction DR2. A corner where the long side in the first direction DR1 and the short side in the second direction DR2 meet may be rounded with a predetermined radius of curvature or may be a right-angle. The shape of the display device **10** in a plan view is not limited to the rectangular shape, and may be a shape similar to other polygonal shapes, a circular shape, or an elliptical shape.

[0040] The display device **10** may include a display panel **100**, a display driver **200**, a circuit board **300**, a touch driver **400**, and a power supply unit **500**.

[0041] The display panel **100** may include a display area DA and a non-display area NDA.

[0042] The display area DA may include a plurality of pixels disposed therein for displaying an image. The display area DA may emit light from a plurality of emission areas or a plurality of opening areas. For example, the display panel **100** may include pixel circuits including switching elements, a pixel defining film defining the emission areas or the opening areas, and self-light emitting elements.

[0043] The non-display area NDA may be disposed around and/or adjacent to the display area DA. The non-display area NDA may be an area outside the display area DA. For example, the non-display area NDA may be an edge area of the display area DA of the display panel **100**. The non-display area NDA may include a gate driver disposed therein supplying gate signals to gate lines, and fan-out lines connecting the display driver **200** and the display area DA to each other. The non-display area NDA may include the display driver **200** and pad portions connected to the circuit board **300**.

[0044] The display driver **200** may output signals and voltages for driving the display panel **100**. The display driver **200** may supply data voltages to data lines. The display driver **200** may supply source voltages to power lines and supply gate control signals to the gate driver. The display driver **200** may be formed as an integrated circuit (IC) and mounted on the display panel **100** in a chip on glass (COG) manner, a chip on plastic (COP) manner, or an ultrasonic bonding manner. As an example, the display driver **200** may be disposed in the non-display area NDA. As another example, the display driver **200** may be mounted on the circuit board **300**.

[0045] The circuit board **300** may be attached onto the pad portions of the display panel **100** using an anisotropic conductive film (ACF). Lead lines of the circuit board **300** may be electrically connected to the pad portions of the display panel **100**. The circuit board **300** may be a flexible printed circuit board, a flexible printed circuit, a printed circuit board, or a flexible film such as a

chip on film.

[0046] The circuit board **300** may include a plurality of slits **700**. For example, the circuit board **300** may include a bonding area that overlaps the pad portions of the display panel **100** and a non-bonding area that does not overlap the pad portions of the display panel **100**, and the plurality of slits **700** may be disposed in the bonding area BA of the circuit board **300**. In a plan view, each slit **700** may have a shape that extends along the second direction DR2. In a plan view, the plurality of slits **700** may be arranged along the first direction DR1. Here, the first direction DR1 may correspond to, for example, an elongation direction of the circuit board **300**. The slit **700** may control an elongation rate of the circuit board **300**. For example, the slit **700** may control an elongation rate of the circuit board **300** in the first direction DR1. To this end, according to an embodiment of the present invention, the slit **700** may extend along a direction (e.g., the second direction DR2) crossing the elongation direction (e.g., the first direction DR1) of the circuit board **300**. Pressure is applied to the circuit board **300** during a process of attaching the circuit board **300** and the display panel **100** to each other, and the circuit board **300** may be elongated (or expanded) along the first direction DR1 by such pressure. Since the display panel **100** is made of a material that is harder than the circuit board **300**, the circuit board **300** may be elongated by the pressure described above, while the display panel **100** might not be elongated by the pressure described above, such that a problem that pad electrodes of the circuit board **300** and pad electrodes of the display panel **100** are not in contact with each other may occur. An elongation amount (or an elongation degree) of the circuit board **300** tends to be gradually accumulated and increased along the first direction DR1 or a direction reverse (or opposite) to the first direction DR1 (hereinafter referred to as a first reverse direction) from a central portion of the circuit board **300**. Accordingly, the circuit board **300** may have a greater elongation amount on both side surfaces thereof than at the central portion thereof. The slit **700** may have a shape in which it cuts a portion of the circuit board **300** in the direction crossing the first direction to prevent the elongation amount from being accumulated in the first direction. An example in which five slits **700** are arranged in the bonding area BA has been illustrated in FIG. 1, but this is only an example, and the number of slits **700** described above is not limited thereto and may be variously changed.

[0047] The touch driver **400** may be mounted on the circuit board **300**. The touch driver **400** may be electrically connected to a touch sensing unit of the display panel **100**. The touch driver **400** may supply touch driving signals to a plurality of touch electrodes of the touch sensing unit and sense change in capacitance between the plurality of touch electrodes. For example, the touch driving signal may be a pulse signal having a predetermined frequency. The touch driver **400** may decide whether or not an input has occurred and calculate input coordinates, based on the change in capacitance between the plurality of touch electrodes. The touch driver **400** may be formed as an integrated circuit (IC).

[0048] The power supply unit **500** may be disposed on the circuit board **300** and may supply source voltages to the display driver **200** and the display panel **100**. The power supply unit **500** may generate a driving voltage and supply the driving voltage to a driving voltage line, may generate an initialization voltage and supply the initialization voltage to an initialization voltage line, may generate a reference voltage and supply the reference voltage to a reference voltage line, and may generate a common voltage and supply the common voltage to a common voltage line. In this case, the common voltage of the common voltage line may be supplied to a common electrode of light emitting elements ED of the plurality of pixels. The driving voltage may be a high potential voltage for driving the light emitting element ED, and the common voltage may be a low potential voltage for driving the light emitting element ED.

[0049] FIG. 2 is an exploded perspective view of a circuit board **300** according to an embodiment of the present invention, FIG. 3 is a cross-sectional view taken along line X1-X1' of FIG. 2, FIG. 4 is a cross-sectional view taken along line X2-X2' of FIG. 2, and FIG. 5 is a cross-sectional view taken along line X3-X3' of FIG. 2. For example, FIG. 2 may be an exploded perspective view

of the circuit board **300** of FIG. **1** described above.

[0051] The circuit board **300** may include an insulating layer **310**, pad electrodes **320**, a base layer **330**, an adhesive layer **340**, and a cover layer **350** as in an example that is illustrated in FIGS. **2** to **5**.

[0052] The insulating layer **310** may be disposed in the non-bonding area NBA. The insulating layer **310** may include a polyimide.

[0053] The pad electrodes **320** may be disposed on the insulating layer **310**. For example, the pad electrodes **320** may be disposed on the insulating layer **310** in the non-bonding area NBA and may be disposed in the bonding area BA. In this case, the pad electrodes **320** may be covered by the insulating layer **310** in the non-bonding area NBA. The pad electrode **320** may include a metal. The circuit board **300** and the pad portions of the display panel **100** may be electrically connected to each other through the pad electrodes **320**.

[0054] The base layer **330** may be disposed on the pad electrodes **320**. For example, the base layer **330** may be disposed on the pad electrodes **320** in the non-bonding area NBA and the bonding area BA. In this case, the base layer **330** may cover the pad electrodes **320** in the non-bonding area NBA and the bonding area BA. The base layer **330** may be made of the same material as the insulating layer **310** described above. For example, the base layer **330** may include a polyimide.

[0055] The adhesive layer **340** may be disposed on the base layer **330**. For example, the adhesive layer **340** may be disposed on the base layer **330** in the non-bonding area NBA and the bonding area BA. In this case, the adhesive layer **340** may overlap a portion of the base layer **330** in the non-bonding area NBA. The adhesive layer **340** may be disposed between the base layer **330** and a cover layer **350** to be described later and may adhere the base layer **330** and the cover layer **350** to each other.

[0056] The cover layer **350** may be disposed on the adhesive layer **340**. For example, the cover layer **350** may be disposed on the adhesive layer **340** in the non-bonding area NBA and the bonding area BA. For example, the cover layer **350** may cover an entirety of adhesive layer **340**. In this case, the cover layer **350** may overlap a portion of the base layer **330** in the non-bonding area NBA. The cover layer **350** may be a reinforcing layer capable of supporting the circuit board **300** that is flexible.

[0057] As in an example illustrated in FIGS. **2**, **4**, and **5**, the slit **700** may be formed in the base layer **330**. For example, the plurality of slits **700** may be disposed in the bonding area BA of the base layer **330**. In a plan view, each slit **700** may have a shape in which it extends along the second direction DR2. In a plan view, the plurality of slits **700** may be arranged along the first direction DR1. Here, the first direction DR1 may correspond to, for example, an elongation direction of the base layer **330** that is provided in the circuit board **300**.

[0058] According to an embodiment of the present invention, as illustrated in FIGS. **2** and **5**, the base layer **330** may include a plurality of sub-bonding portions SB that are divided by the plurality of slits **700** in the bonding area BA. The plurality of sub-bonding portions SB may be disposed in the bonding area BA.

[0059] Depending on the elongation rate of the circuit board **300**, a predetermined length L of the sub-bonding portion SB (or a distance between adjacent slits **700**) for connecting the circuit board **300** and the display panel **100** to each other may be changed.

[0060] The length L of the sub-bonding portion SB (or the distance between the adjacent slits **700**) may be determined by the following equation.

[0061] When the length L of any one sub-bonding portion SB is defined as “C”, a maximum elongation amount of any one sub-bonding portion SB is defined as “B”, and an elongation rate of any one sub-bonding portion SB per unit length is defined as “A”, the length of the sub-bonding portion SB (or the distance between the adjacent slits **700**) described above may be determined by the following equation:

$$C=B/A.$$

[0062] Here, A may refer to an elongation rate of the circuit board **300** per unit length. For example, A may refer to an elongation rate of the base layer **330**, which is included in the circuit board **300**, per unit length. The elongation rate of the base layer **330** per unit length is a unique numerical value that is determined depending on material properties or the like of the base layer **330**, and is a numerical value known in advance.

[0063] B is the maximum elongation amount, which may be related to an overlap ratio between an outermost pad electrode **320** (hereinafter referred to as a first outermost pad electrode **320**) of the circuit board **300** and an outermost pad electrode (hereinafter a second outermost pad electrode) of the display panel **100** that corresponds to the first outermost pad electrode **320**. Here, the first outermost pad electrode **320** may refer to a pad electrode **320** disposed on the outermost portion of one side of the sub-bonding portion SB among a plurality of pad electrodes **320** disposed on the sub-bonding portion SB. According to an embodiment of the present invention, B may be smaller than or equal to 50 (or 50%). For example, B may be set within 50% of a pitch PT of the first outermost pad electrode **320**. However, a numerical value of B is not limited thereto, and may be variously changed.

[0064] An example of calculating the length of the sub-bonding portion SB (e.g., the distance between the adjacent slits **700**) described above will be described in detail below.

[0065] For example, when the elongation rate A of the base layer **330** (e.g., polyimide) of the circuit board **300** per unit length is 1 $\mu\text{m}/1\text{ mm}$ and the maximum elongation amount B is 50, the length C of the sub-bonding portion SB of the circuit board **300** may be 50 mm. For example, the distance between the adjacent slits **700** of the circuit board **300** may be 50 mm. Here, the maximum elongation amount B may be set as an occupancy ratio of the pad electrode **320** when the pitch PT (e.g., a cycle) of the first outermost pad electrode **320** is 100 and an occupancy ratio and a non-occupancy ratio of the pad electrode **320** at the pitch PT of 100 are 50:50, as in the example illustrated in FIG. 5.

[0066] According to an embodiment of the present invention, the respective sub-bonding portions SB of the circuit board **300** may have the same length as each other. For example, the plurality of slits **700** may be disposed at equal intervals. For example, each sub-bonding portion SB may have a length of about 50 mm.

[0067] FIGS. 6 to 8 are cross-sectional views for describing processes of a method for fabricating the display device according to an embodiment of the present invention.

[0068] First, as illustrated in FIG. 6, a substrate **800** of a display panel **100** on which pad electrodes **820** are disposed is prepared, and an anisotropic conductive film **900** including a plurality of conductive particles **910** may be disposed on the pad electrode **820** and the substrate **800**.

Thereafter, the circuit board **300** having the plurality of slits **700** described above may be disposed on the anisotropic conductive film **900**.

[0069] According to an embodiment of the present invention, in consideration of elongation of the circuit board **300** during a pressurizing process to be described later, an interval between the pad electrodes **320** (hereinafter referred to as first pad electrodes **320**) of the circuit board **300** may be different from an interval that is between the pad electrodes **820** (hereinafter referred to as second pad electrodes **820**) of the display panel **100**. For example, the interval that is between the first pad electrodes **320** may be smaller than the interval that is between the second pad electrodes **820**.

[0070] Next, as illustrated in FIG. 7, a heating/pressurizing header **160** may be disposed on the circuit board **300**. According to an embodiment of the present invention, a heat conduction sheet **150** may be disposed between the circuit board **300** and the heating/pressurizing header **160**. The heat conduction sheet **150** may evenly transfer heat and pressure of the heating/pressurizing header **160** to the circuit board **300** that is located below the heat conduction sheet **150**. For example, the heat conduction sheet **150** may include a TeflonTM material.

[0071] Next, as illustrated in FIG. 8, the heating/pressurizing header **160** may descend along a

direction of an arrow AR1 to pressurize the circuit board **300**. For example, the heating/pressurizing header **160** may cure the anisotropic conductive film **900** that is disposed between the circuit board **300** and the substrate **800** of the display panel **100** by heating and pressurizing the heat conduction sheet **150**. In addition, in such a pressurizing process, the first pad electrodes **320** and the second pad electrodes **820**, which correspond to each other, may be electrically connected to each other by the conductive particles **910** that are disposed between the first pad electrodes **320** and the second pad electrodes **820**. In this case, the base layer **330** of the circuit board **300** may be elongated (or expanded) in directions (e.g., directions of a dotted line arrow AR2 or the first direction DR1 and the first reverse direction) perpendicular to a pressurizing direction (e.g., a direction reverse to a third direction DR3 or the direction of the arrow AR1) by the pressure through the heating/pressurizing header **160**.

[0072] Due to the elongation of the base layer **330**, the interval between the first pad electrodes **320** that are disposed on the base layer **330** increases, such that the first pad electrodes **320** and the second pad electrodes **820** may be aligned with each other. In other words, due to the elongation of the base layer **330**, the first pad electrodes **320** and the second pad electrodes **820** corresponding to each other may be disposed to overlap each other. In addition, a width of the slit **700** of the base layer **330** may be reduced due to the elongation of the base layer **330**. For example, referring to FIGS. **6** and **8**, a width W2 of the slit **700** of the base layer **330** after being elongated may be smaller than a width W1 of the slit **700** of the base layer **330** before the elongation of the base layer **330**. In other words, each of the sub-bonding portions SB of the base layer **330** is elongated due to the elongation of the base layer **330**, and accordingly, a distance between adjacent sub-bonding portions SB (e.g., the width of the slit) may be reduced.

[0073] With the display device according to an embodiment of the present invention, the base layer **330** of the circuit board **300** includes the plurality of slits **700**, and accordingly, the elongation amount of the circuit board **300** may be reduced. Accordingly, alignment accuracy between the pad electrodes **320** of the circuit board **300** and the pad electrodes **820** of the display panel **100** that correspond to the pad electrodes **320** may be increased.

[0074] In addition, with the display device according to an embodiment of the present invention, an accurate length of the sub-bonding portion SB (or an accurate distance between the adjacent slits **700**) may be calculated based on the elongation rate of the base layer **330**, and thus, the alignment accuracy between the pad electrodes **320** of the circuit board **300** and the pad electrodes **820** of the display panel **100** that corresponds to the pad electrodes **320** may be further increased.

[0075] While the present invention has been described with reference to the embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and detail may be made thereto without departing from the spirit and scope of the present invention.

Claims

1. A display device comprising: a display panel; and a circuit board connected to the display panel, wherein the circuit board includes a plurality of sub-bonding portions that are separated from each by a plurality of slits, and a length of each of the plurality of sub-bonding portions is set based on an elongation rate of the circuit board.
2. The display device of claim 1, wherein the length of the sub-bonding portion is calculated by a following equation:
$$C=B/A$$
, where A is an elongation rate of the sub-bonding portion per unit length, B is a maximum elongation amount of the sub-bonding portion, and C is the length of the sub-bonding portion.
3. The display device of claim 2, wherein B is set within 50% of a pitch of an outermost pad electrode of a plurality of pad electrodes that are disposed on the sub-bonding portion.
4. The display device of claim 1, wherein the circuit board includes: an insulating layer; a pad electrode disposed on the insulating layer; a base layer disposed on the pad electrode; an adhesive

- layer disposed on the base layer; and a cover layer disposed on the adhesive layer.
5. The display device of claim 4, wherein the base layer includes the plurality of slits.
6. The display device of claim 5, wherein the base layer includes the plurality of slits in a bonding area of the circuit board.
7. The display device of claim 5, wherein the plurality of slits are arranged in the base layer along a first direction, each of the plurality of slits extends in a second direction, and the first direction and the second direction cross each other.
8. The display device of claim 5, wherein the base layer includes a polyimide.
9. The display device of claim 6, wherein the insulating layer includes a same material as the base layer.
10. The display device of claim 1, further comprising an anisotropic conductive film disposed between the display panel and the circuit board.
11. The display device of claim 5, wherein the insulating layer is disposed in a non-bonding area of the circuit board.
12. The display device of claim 5, wherein the pad electrode is disposed in a non-bonding area and a bonding area of the circuit board.
13. The display device of claim 5, wherein the base layer is disposed in a non-bonding area and a bonding area of the circuit board.
14. The display device of claim 5, wherein the adhesive layer is disposed in a non-bonding area and a bonding area of the circuit board.
15. The display device of claim 5, wherein the cover layer is disposed in a non-bonding area and a bonding area of the circuit board.
16. The display device of claim 5, wherein the pad electrode of the circuit board is connected to a pad electrode of the display panel.
17. The display device of claim 16, wherein the pad electrode of the circuit board is connected to the pad electrode of the display panel in a bonding area of the circuit board.
18. A method for fabricating a display device, comprising: disposing an anisotropic conductive film on a substrate of a display panel; disposing a circuit board on the anisotropic conductive film, wherein the circuit board includes a plurality of sub-bonding portions that are separated from each other by a plurality of slits; disposing a heat conduction sheet on the circuit board; disposing a heating/pressurizing header on the heat conduction sheet; and connecting a pad electrode of the circuit board and a pad electrode that is disposed on the substrate to each other by pressurizing and heating the heat conduction sheet with the heating/pressurizing header, wherein a length of each of the plurality of sub-bonding portions is set based on an elongation rate of the circuit board.
19. The method for fabricating a display device of claim 18, wherein the length of the sub-bonding portion is calculated by a following equation:
$$C=B/A$$
where A is an elongation rate of the sub-bonding portion per unit length, B is a maximum elongation amount of the sub-bonding portion, and C is the length of the sub-bonding portion.
20. The method for fabricating a display device of claim 19, wherein B is set within 50% of a pitch of an outermost pad electrode of a plurality of pad electrodes that are disposed on the sub-bonding portion, wherein the plurality of pad electrodes includes the pad electrode of the circuit board.
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